

A proposed model to test the hypothesis of exercise-induced localized fat reduction (spot reduction), including a systematic review with meta-analysis

Short title: Exercise-induced spot reduction

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ABSTRACT

Background: The notion that specific exercises reduce localized adipose tissue depots (i.e., targeted fat loss) and modify fat distribution is commonly termed spot reduction. According to this long-held popular belief, exercising a limb would lead to greater reduction in the adjacent adipose tissue in comparison to the contralateral limb. Aside from popular wisdom, scientific evidence from the 20th and 21th century seems to offer inconclusive results. **Objective:** To summarize the peer-reviewed literature assessing the effects of unilateral limb training, compared to the contralateral limb, on the localized adipose tissue depots on healthy participants, and to meta-analyse its results. **Methods:** We followed the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). We searched PubMed, Web of Science, and SCOPUS electronic databases, using several relevant keywords combinations. Independent experts were contacted to help identify additional relevant articles. Following a PICOS approach, we included controlled studies that incorporated a localized exercise intervention (i.e., single-limb training) to cohorts of healthy participants (i.e., no restriction for fitness, age, or sex), compared to a control condition (i.e., contralateral limb), where the main outcome was the pre-to-post intervention change of localized fat. The methodological quality of the studies was assessed using the Physiotherapy Evidence Database scale. Pre- and post-intervention mean $\pm$ standard deviation for fat-related outcome from the trained and control groups (limbs) were converted to Hedges' g effect size (ES; with 95% confidence intervals [CI]), using a random-effects model. The impact of heterogeneity was assessed using the I^2 statistic. The risk of reporting bias was explored using the extended Egger's test. The statistical significance threshold was set at $p < 0.05$. **Results:** From 1,833 search records initially identified, 13 were included in the meta-analysis, involving 1,158 male and female participants (age, 14-71 years). The 13 studies achieved a high methodological quality, and results with low heterogeneity ($I^2 = 24.3\%$) and no bias (Egger's test $p = 0.133$). The meta-analysis involved 37 comparisons, with 17 of these favouring (i.e., greater reduction of localized fat) the trained limb, and 20 favouring the untrained limb, but the ES ranged between -1.21 to 1.07. The effects were consistent, with a pooled ES = -0.03, 95% CI -0.10 to 0.05, $p = 0.508$, meaning that spot reduction was not observed. **Conclusion:** Localized muscle training has no effect on localized adipose tissue depots, i.e., no spot reduction, regardless of the characteristics of the population and of the exercise program. The popular belief on spot reduction is probably derived from wishful thinking, and convenient marketing strategies, such as influencers seeking increased popularity and procedures' sellers interested in increasing advertising. **Keywords:** exercise; human physical conditioning; resistance training; high-intensity interval training; body composition; subcutaneous fat.

1. INTRODUCTION

“A man may box and fence, and even walk, without losing his terrible abdominal accumulation; but if he centres his efforts at muscular exertion on the abdomen itself the fat cannot stand the attack and will gradually disappear”.
Checkley, E. (1895) [1].

Since (at least) the 19th century, the notion that specific exercises can reduce localized adipose tissue depots (i.e., targeted fat loss) and modify fat distribution remains a very popular belief, commonly termed spot reduction [2]. From the mid to nearly the end of the 20th century, several studies were performed on the subject, suggesting that spot reduction might be feasible [3-6]. However, during the same period several studies disproved the notion of spot reduction [2, 7-13]. Toward the end of the 20th century there seemed to be a consensus among the scientific community that spot reduction is a myth. Nonetheless, during the 21th century new studies [14-24] relaunched the debate.

Why is the notion of spot reduction so appealing across centuries [1, 25, 26]? Why have researchers not reached a definitive answer to the problem? This might be explained by three main factors. A first factor may be the difficulty inherent to address the hypothesis of spot reduction. There are complex interactions between i) different exercise programming characteristics (e.g., exercise modality; periodization; load management; adherence to the program), ii) differential regional responses of adipose tissue depots to exercise (i.e., lipolysis; re-esterification; mobilization of free fatty acids), and iii) inter-individual differences in the modulators of the fat metabolism in response to exercise (e.g., sex; obesity) [19, 27-32]. A second factor of controversy may arise from different concepts of spot reduction [16, 33]. Different models of study were used to test the hypothesis of spot reduction, such as cross-sectional studies [7, 14, 34, 35], as well as long-term intervention studies involving exercise compared to nutrition [22, 23], trunk-localized exercise [8, 20], limb-localized exercise [5, 9, 18, 21], and whole-body exercise [15, 16]. A third factor is the difficulty to conduct rigorous experimentation to test such a hypothesis (e.g., control the participants' diet and their compliance to the program; use valid measurement techniques [18]). The difficulty encountered by scientists is in contrast with the ease with which personal beliefs (or publicity) can be communicated [1, 25, 26]. Marketing and science often collide [33], and marketing the notion of exercise-based spot reduction to persons seeking a desperate solution to their problems [36] may be very appealing.

If the notion of spot reduction is correct, then performing a regimen of unilateral exercise should lead to higher reduction in adipose content in that region than that in the contralateral limb. To the best of our knowledge, the debate regarding exercise-based spot reduction seems to be active, even after (at least) three centuries [1]. To contribute to settle down the debate, a systematic review with meta-analysis was conducted to qualitatively assess and quantitatively summarize the evidence in the field, but also circumvent the problem of most exercise-sport related studies: a reduced sample size [37]. Our aim was to summarize the peer-reviewed literature assessing the effects of unilateral limb training, compared to the contralateral limb, on the localized adipose tissue depots on healthy participants across the life span, and to meta-analyse its results.

2. METHODS

We followed the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [38, 39]. The methods were established before initiating the research, and protocol registration preceded the search.

2.1. Search strategy

We searched through PubMed, Web of Science, and SCOPUS electronic databases from the inception of indexing to June 2021, with no restriction on language. Potentially relevant keywords were collected through authors' consensus based on previous studies conducted in relation to spot reduction, and organized vocabulary (i.e., Medical Subject Headings: MeSH) was also incorporated. As a result, the following key words were introduced in the electronic databases in different combinations using Boolean search syntax with the operators "AND", "OR": activity, arm, body, clinical, composition, conditioning, controlled, distribution, dominant, elbow, exercise, extension, fat, flexion, forearm, high, human, intensity, interval, knee, leg, local, localized, loss, mass, modalities, model, motor, movement, muscle, musculoskeletal, non-dominant, phenomena, physical, physiological, reduction, regional, resistance, running, single, sport, spot, strength, subcutaneous, targeted, therapy, thigh, training, treatment, trial, unilateral. Electronic searches were conducted according to the specific characteristics of each electronic database search engine. For example, in the PubMed database, the following search syntax was used: controlled clinical trial [Publication Type] AND training [Title/Abstract] OR single-leg [Title/Abstract] AND body composition [MeSH Terms] AND fat [Title/Abstract].

After the initial search in June 2021, we created accounts in the respective databases. Through these accounts, the lead investigator received weekly automatically generated emails for updates regarding the search terms

used (if available). All studies that were published before August 2021 were considered for inclusion. We excluded studies based on the review of the title, abstract, or (when needed) after the full text was read. Conference proceedings were considered if the full-text was available. The reference list of included studies was searched for potentially relevant studies. Two authors (RRC, DCA) conducted the process independently, with potential discrepancies resolved by consensus.

Thereafter, the list of included articles and the inclusion criteria were sent to two independent world experts in the field of body composition (<https://www.expertscape.com/ex/body+composition>) to help identify additional relevant articles. Additionally, the experts (i) hold a Ph.D. in Sports Sciences or related field (e.g., Health Sciences); (ii) have peer-reviewed publications in body composition, in journals with impact factor according to the Journal Citation Reports[®]. The experts were not provided with our search strategy, to avoid biasing their own searches. Upon completion of all these steps, the databases were again consulted in search for errata or retractions of any included study.

2.2. Eligibility criteria

To elaborate the PICO's eligibility criteria, we first elaborated a definition for the investigated problem. Accordingly, spot reduction (in humans) is defined as a greater reduction of the non-intramuscular fat-related depot(s) (e.g., subcutaneous fat) adjacent to a voluntary exercised muscle compared to the same depot from the contralateral non-exercised muscle, after an intervention period.

Accordingly, and following PICOS criteria, we incorporated studies that: (i) included cohorts of healthy (e.g., with medical/ERB clearance to participate in a training programme) participants (humans), with no restriction for fitness/sport background, age or sex. Excluded participants were those with physical trauma (e.g., limb amputation) [40], health diseases (e.g., stroke leading to paretic limb [41]; genetic conditions or syndromes potentially affecting adipose tissue or its response to training [24, 42, 43]); (ii) incorporated a localized exercise intervention (without restriction from the mode of exercise, e.g., resistance training; endurance training) where one limb was trained and contralateral limb was the control. Interventions with a minimum of 2 weeks were considered [44, 45]. Studies that incorporated a non-localized exercise intervention (e.g., running; bilateral leg press) were excluded. Cross-sectional studies were also excluded. Studies were not excluded if they lacked dietary control and/or included nutritional supplementation, as this is not a critical factor for experimental models using a contralateral limb as a control condition [46]; (iii) localized exercise was compared with a control condition (i.e., contralateral limb), with the only difference between the conditions being the exercise

intervention; (iv) the study included a pre-to-post intervention assessment of at least one fat-related parameter (e.g., fat mass; fat volume) using DEXA, magnetic resonance imaging, computerized tomography, skinfold callipers, ultrasound, and microscopic method (i.e., subcutaneous fat biopsy). Secondary outcomes were considered, including potential adverse effects derived from the intervention (e.g., injury); and (v) utilized a randomised or non-randomised controlled design, as long as at least one comparator group existed.

2.3. Data extraction

Two authors of the review (RRC, DCA) performed the data extraction independently, using a pre-defined form created in Microsoft Excel (Microsoft Corporation, Redmond, WA, USA). If there were any discrepancies between the authors in the extracted data, the accuracy of the information was re-checked in the studies. We extracted the following data: participants' sex, age (years), body mass (kg), height (m), and previous experience with training. If applicable, information about the type and level (e.g., professional, amateur) of sport practice was also retrieved. Regarding training characteristics, extracted data included training frequency (days/week) and training duration (weeks), intensity level and marker of intensity (e.g., % of one repetition maximum [1RM]), total volume (e.g., repetitions; minutes), types of exercises performed, combination of exercise with diet, and progressive overload techniques (if any).

The means and standard deviation of dependent variables were extracted at pre- and post-intervention time points from included studies. In cases where the required data were not clearly or completely reported, the authors of the study were contacted for clarification. If no response was obtained from the authors (after two attempts), or if the authors could not provide the requested data, the study outcome was excluded from the analysis. However, even when no numerical data were provided by the authors upon contact, in cases where data were displayed in a figure [9], the meta-analysis used validated ($r = 0.99$, $p < 0.001$) [47] software (WebPlotDigitizer; <https://apps.automeris.io/wpd/>) [48] to derive the relevant numerical data.

2.4. Methodological quality assessment

The Physiotherapy Evidence Database (PEDro) scale was used to assess the methodological quality of the included studies [49]. There are 11 items on the PEDro checklist, but item 1 is not included in the total score. Therefore, the methodological quality of the included studies was rated from 0 (lowest quality) to 10 (highest quality). This scale evaluates different aspects of the study design, such as participant eligibility criteria, randomization, blinding, attrition, and reporting of data. The validity and reliability of the PEDro scale has

been established previously [49-51]. Additionally, its agreement with other scales (e.g., Cochrane risk of bias tool) has been reported [52]. Moreover, the PEDro scale is probably one of the most frequently used scales in the literature, helping to make comparisons between meta-analyses. According to cut-off scores, the methodological quality was rated as ‘poor’ (<4), ‘fair’ (4-5), ‘good’ (6-8) and ‘excellent’ (9-10) in some sub-fields, however, it is not possible to satisfy all scale items in some areas of physiotherapy practice [53]. Moreover, in the context of this study, and according to the definition of spot reduction and the proposed experimental model to test the hypothesis of spot reduction, is not possible to blind the participants regarding whether they trained or not one of their limbs, making the item 5 from the PEDro scale an unfair criteria to assess the methodological quality of studies in the context of our review. Therefore, as outlined in previous systematic reviews in some sub-fields of physiotherapy [54, 55], the methodological quality of studies was interpreted using the following convention, based on the summary score: studies that scored ≤ 3 points were considered as being of “poor quality”, studies scoring 4 or 5 points were considered as being of “moderate quality”, and studies that scored 6 – 10 points were considered as being of “high quality”. Two authors (RRC, DCA) performed the methodological quality assessment independently. Disagreements in the assessments between the reviewers were resolved through discussion and consensus.

2.5. Statistical analysis

Pre- and post-intervention mean $\pm$ standard deviation (SD) for a given fat-related outcome from the trained and control groups were converted to Hedges’ g effect size (ES). A meta-analysis for a given fat-related outcome was conducted if at least three studies provided sufficient data for the calculation of ES [56-58]. The data were standardized using post score SD. For studies that reported standard errors, standard deviations were calculated by multiplying the standard error with the square root of the sample size [59]. In all analyses, we used the random-effects model to account for differences between studies that might affect the treatment effect [60, 61]. The ES values are presented alongside their respective 95% CIs. Calculated ES were interpreted using the following scale: < 0.2, trivial; 0.2 – 0.6, small; > 0.6 – 1.2, moderate; > 1.2 – 2.0, large; > 2.0 – 4.0, very large; > 4.0, extremely large [62]. The impact of heterogeneity was assessed using the I^2 statistic, with values of < 25%, 25 - 75%, and > 75% considered to represent low, moderate, and high levels of heterogeneity, respectively. The risk of reporting bias was explored using the extended (two-tailed) Egger’s test [63]. To adjust for publication bias, a sensitivity analysis was conducted using the trim and fill method (Duval and Tweedie, 2000), with L0 as the default estimator for the number of missing studies (Shi and Lin, 2019). All analyses were carried out using the Comprehensive Meta-Analysis program (version 2; Biostat, Englewood, NJ, USA). The statistical significance threshold was set at $p < 0.05$.

3. RESULTS

3.1. Study selection

A total of 1,833 search records were initially identified. After excluding the duplicates and studies based on title, abstract, 83 studies remained, and the full text was read. From these, 13 were included in the meta-analysis [5, 9, 10, 18, 21, 46, 64-70]. Figure 1 provides a diagram of the study selection process. The included studies involved 1,158 participants (acting as both experimental and control groups). The characteristics of the participants from the included studies, the programming parameters of the training interventions, and the fat-related outcomes (for both the control and experimental limbs) are presented in Table 1.

Figure 1

Table 1

Briefly, training interventions were applied during 2 up to 20 weeks, with a training frequency of 3 sessions per week, up to 7 sessions per week (i.e., daily training). The training intensity (i.e., single-limb) varied from 10% up to 90% of 1RM for those interventions that applied resistance training exercises, and ~40% of peak oxygen consumption (VO_{2peak}) in the intervention that applied endurance (i.e., cycling) training. Of note, the interventions that applied resistance training exercises commonly used elbow flexors- extensors-related exercises (e.g., dumbbell bicep concentration curls; overhead triceps extension), or knee extensors-related exercises (e.g., seated leg press; seated leg extension), although none of the included studies applied knee flexors-related exercises. No major adverse effects were reported among the included studies, other than mild-moderate delayed onset of muscle soreness. However, most of the included studies in this meta-analysis failed to report specific information regarding adverse health effects. This reflects a larger problem in sports sciences and produces unbalanced accounts, as authors report the main effects, but not the potential adverse health effects.

3.2. Methodological quality

Using the PEDro checklist, the 13 studies achieved 6-8 points and were classified as being of “high” methodological quality (Table 2).

*****Table 2*****

3.3. Meta-analysis results

The meta-analysis included 13 controlled studies, involving 37 comparisons, with 17 of these favouring (i.e., greater reduction of localized fat) the trained limb, and 20 favouring the untrained limb, but the ES ranged between -1.21 to 1.07. The effects were consistent, with a pooled ES = -0.03, 95% CI -0.10 to 0.05, $p = 0.508$, $I^2 = 24.3\%$, Egger's test $p = 0.133$ (Figure 2), meaning that spot reduction was not observed.

*****Figure 2*****

4. DISCUSSION

According to the definition for our proposed model to test the hypothesis of spot reduction, our aim was to summarize the peer-reviewed literature assessing the effects of unilateral limb training, compared to the contralateral limb, on the localized adipose tissue depots on healthy participants across the life span, and to meta-analyse its results. From the 13 studies included in our meta-analysis, all achieved six or more points in the PEDro scale. This may increase the perceived quality of research included in our analyses and the confidence in evidence. Further, results were obtained with low impact of heterogeneity ($I^2 = 24.3\%$) and no significant risk of reporting bias (Egger's test $p = 0.133$). In addition, a total of 1,158 participants were included in the 13 studies, a strength when compared to the relatively reduced number of participants involved in sport sciences literature [37]. Although exercise is a potent contributor to fat reduction [71], our meta-analysis indicated no significant (trivial) effect of localized muscle training on localized adipose tissue depots, i.e., no spot reduction was observed. Therefore, long-term exercise-based localized adipose tissue reduction would not be an expected result from an adequately planned exercise intervention. The result from our meta-analysis is based on interventions with a mean duration of 11 weeks (range, 2 to 20 weeks), involving different training approaches (e.g., cycling; resistance training), in participants of different sex, age, and physical fitness level (e.g., sedentary; physically active). Despite the heterogeneity in sample, protocols, and study designs, the lack of an effect was consistent, denoting a robust phenomenon that is largely independent of the characteristics of the population or of the exercise program. It is indeed intriguing from a physiological and anatomical perspective how exercise-based interventions may induce a localized effect on skeletal muscle tissue [72], bone tissue [73], or even skin tissue [74], but not on adipose tissue.

Such an intriguing phenomenon generated controversy since (at least) the 19th century [1, 25, 26], with several studies performed on the subject from the mid of the 20th century up to recently [2-24]. It is possible that the controversy regarding spot reduction relates to its definition. For example, if spot reduction considers the intra-muscular fat stores, a localized reduction may occur, contrary to the subcutaneous fat depot [24]. According to our definition of voluntary exercise-based localized fat reduction (i.e., spot reduction, see methods section, sub-section eligibility criteria), a valid model to test the hypothesis of spot reduction would be one in which, essentially, the muscles in one part of the body are trained, whereas the muscles in the contralateral side are not. Indeed, the use of an appropriate research model is fundamental to researchers avoid flawed experiments that may lead them to inappropriate (or even intended) results. For such definition and proposed model we considered [19, 27-30, 32, 69] i) fat depots from different body regions are not equally comparable within a given individual (i.e. comparing arms and legs); ii) for the same body fat depot, significant inter-individual differences might occur. For example, abdominal fat may respond differently to exercise in male compared to female [28, 75, 76]; iii) contrary to neuromuscular-related outcomes, there is no evidence for a *cross-education* between subcutaneous fat depots through exercise; iv) the effects of exercise training on one limb compared to the contralateral non-exercised limb allow a tight control for dietary (even if this is not manipulated) and other possible intervening factors (e.g., methodology; seasonal variation; genetic; biology; variations in attention and motivation between experimental and control groups) [46, 69]; v) studies seeking to validly test the hypothesis of spot reduction should consider the size of the adipose tissue depots adjacent to the trained and respective non-trained muscles before and after an intervention period (with a relatively high volume of work to impact fat tissue) not just after an acute exercise bout [14]; vi) valid studies should use valid measurement techniques, avoiding techniques that may provide biased results due to changes in muscle mass [18] or other factors not related to biological changes in fat content [77]. For example, reductions between 3-14% (mean 7.5%) were noted in the trained arm compared to the non-trained arm when subcutaneous fat was measured in the biceps using a skinfold calliper [18]. In contrast, when MRI was used to measure arm subcutaneous fat volume, the reduction was nearly three-fold lower (range 0-7%; mean 2.8%) [18]. Additionally, valid studies should report the reliability of measurement (e.g., coefficient of variation; total error of measurement), as not all studies in this field have reported such essential element [4-6, 15].

In contrast to our proposed definition and model to test the hypothesis of exercise-induced localized fat reduction, two cross-sectional studies [14, 17] found acute localized lipolysis. However, the studies did not demonstrate spot reduction (i.e., localized reduction of adipose tissue). Moreover, the two aforementioned cross-sectional studies [14, 17], although found that exercising one leg promoted an increase in lipolysis in the subcutaneous fat adjacent the muscles being exercised (e.g., anterior thigh), the effect was highly local,

meaning that any significant long-term effect (i.e., fat reduction) would be unlikely. Further, compared to the aforementioned cross-sectional studies [14, 17], some authors have found contradicting findings, with intense exercise (e.g., resistance training) reducing subcutaneous adipose tissue blood flow and lipolysis [31]. Aside the controversial findings, the fact that an acute increase in lipolysis does not translate in chronic reduction in fat depots, is analogue to the fact that exercise at a given intensity may allow maximal acute rate of fat oxidation (i.e., FATmax) [78], without long-term effect on body composition [79]. Indeed, even if acute localized lipolysis occurs during exercise, several additional physiological processes are needed before free fatty acids enter the blood stream for later oxidation in tissues [28, 29, 80]. Moreover, the authors from one of the aforementioned cross-sectional studies [14] indicated that *“More calories are expended during aerobic, whole body exercise than by exercise with local muscle groups, and, accordingly, a person seeking to loose fat must be advised to perform whole body exercise”*. Indeed, high intensity exercise have been found to promote large reduction in body fat in different body parts, with many different activities [81]. From a practical point of view, if the main aim of a training program were to improve body composition, including reductions of adipose tissue, the most logically defendable approach would be to include a training programme allowing a considerable energy expenditure density. To this aim, compared to localized exercise, non-localized exercise, involving large muscles groups, would be preferable. Of course, localized exercise may still offer important practical relevance, such as to improve the endurance of trunk muscles (e.g., abdominal muscle training), to induce a cross-education effect on injury limbs, to improve localized-peripheral adaptations with a minimization of central responses (e.g., blood pressure), among others. But the current literature does not support its use for regional fat reduction.

4.1. Limitations

According to our definition, a valid model to test the hypothesis of spot reduction would be one in which the muscles in one limb are trained, whereas the muscles in the contralateral limb are not. To our knowledge, this model is less prone to bias compared to the rest of the models (e.g., cross-sectional; exercise compared to nutrition; trunk-localized exercise; whole-body exercise) currently proposed in the scientific literature to test the hypothesis of spot reduction through exercise training. According to our proposed definition, and model, we conducted a systematic review with meta-analysis that included studies with participants across a wide range of ages, with no restriction to sex or training status, and that included different protocols (e.g., training; assessment techniques). Due to the high heterogeneity between included studies, a high heterogeneity in results might be expected. However, the meta-analysis clearly denotes that lack of spot reduction is ubiquitous, i.e.,

the effect is very strong and seems to be sample-independent and protocol-independent. Although our results seem highly consistent, we discuss some potential limitations.

Firstly, exercising one limb might induce a partial activation of the contralateral limb [82], and contralateral strength gains have been reported [83-85]. How much activation of the control limb might have occurred and to what extent this affected study outcomes is unclear. Additionally, studies usually controlled for the correct technical execution of training exercise by proper spotters and researchers. Therefore, it is assumed that participants recruited for exercise interventions have an adequate exercise technique and supervision that made them able to activate the target muscle while maintaining the contralateral muscle relatively inactive. Secondly, the lack of nutritional control was not considered as an exclusion criterion in our meta-analysis. Nonetheless, the effects of exercise training on one limb compared to the contralateral non-exercised limb allow a tight control for dietary (even if this is not manipulated) and other possible intervening factors (e.g., seasonal variation; genetic; biology) [46, 69]. Thirdly, we only considered voluntary training protocols in this meta-analysis. Therefore, non-voluntary muscle activation strategies and their potential to affect the trained limb [86, 87] were not considered. Fourthly, the studies included in our meta-analysis consisted of training programmes between 2-20 weeks of duration. Therefore, longer-term interventions were not addressed. However, based on current findings, and those derived from some cross-sectional studies involving athletes with several years of training using one limb more than the contralateral limb (e.g., tennis) [7, 35, 88], longer-term interventions would probably help to confirm current findings.

4.2. Conclusions

Localized muscle training has no effect on localized adipose tissue depots, i.e., no spot reduction, regardless of the characteristics of the population and of the exercise program.

4.3. Other information

The protocol for this systematic review with meta-analysis was registered with the International Platform of Registered Systematic Review and Meta-Analysis Protocols (INPLASY) on 28 June 2021 (registration number: INPLASY202160103).

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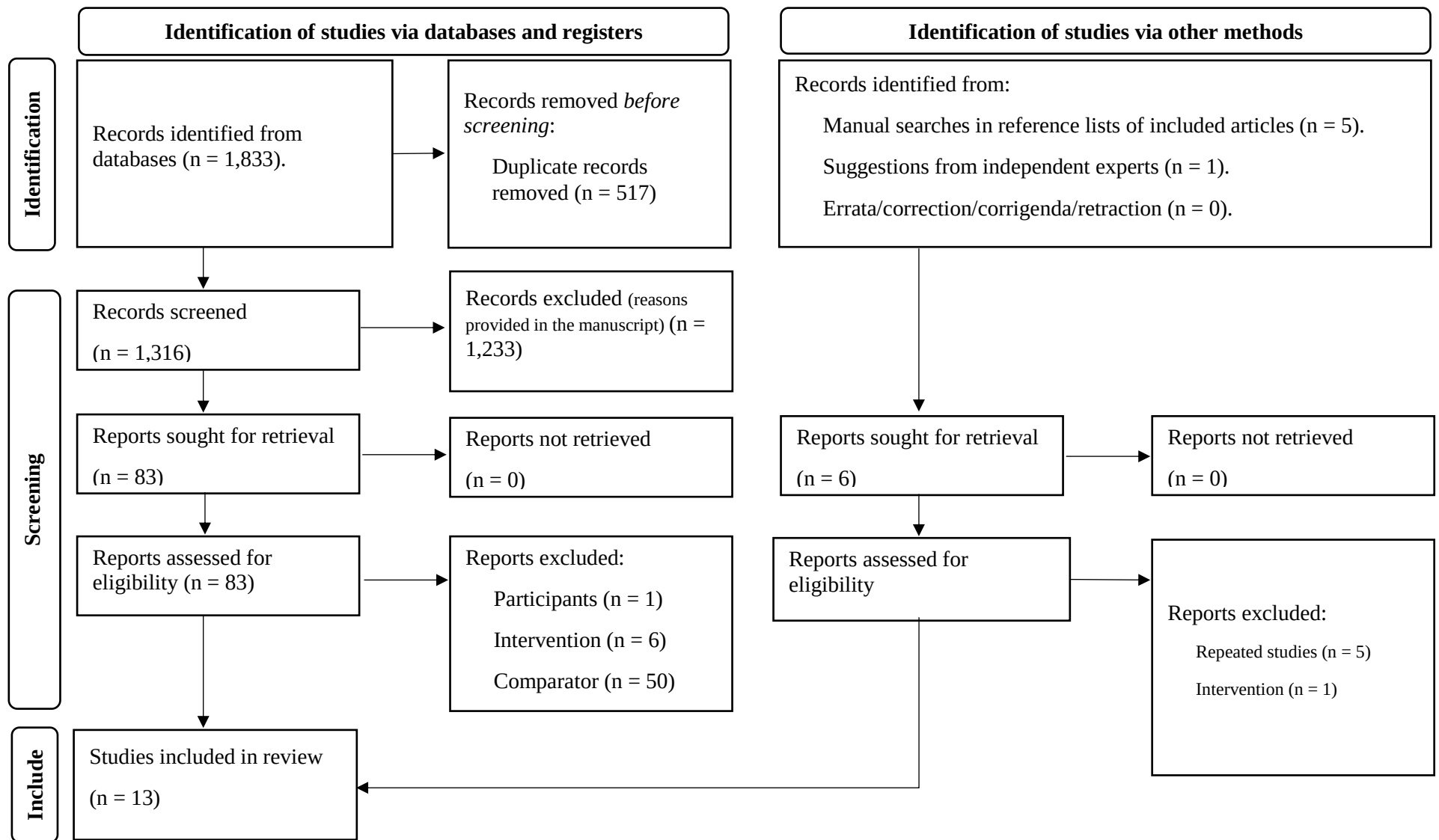


Figure 1. PRISMA 2020 flow diagram.

Study nameStatistics for each studyHedges's g and 95% CI

	Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Brinkworth et al., 2004, bovine colostrum	0.044	0.335	0.112	-0.613	0.700	0.130	0.896
Brinkworth et al., 2004, whey protein	-0.108	0.335	0.112	-0.765	0.549	-0.321	0.748
Devries et al., 2015	0.045	0.255	0.065	-0.454	0.545	0.177	0.859
Hanson et al., 2009 (all) subcutaneous	-0.222	0.205	0.042	-0.625	0.180	-1.083	0.279
Hanson et al., 2009 (men) subcutaneous	1.072	0.317	0.101	0.450	1.694	3.378	0.001
Hanson et al., 2009 (women) subcutaneous	-1.209	0.304	0.092	-1.804	-0.614	-3.982	0.000
Hanson et al., 2009 (all) intermuscular fat	-0.242	0.205	0.042	-0.644	0.161	-1.176	0.239
Hanson et al., 2009 (men) intermuscular fat	0.255	0.297	0.088	-0.328	0.838	0.857	0.391
Hanson et al., 2009 (women) intermuscular fat	-0.240	0.279	0.078	-0.788	0.307	-0.860	0.390
Kostek et al., 2007 (men and women) biceps	-0.096	0.149	0.022	-0.387	0.196	-0.644	0.520
Kostek et al., 2007 (men) biceps	-0.216	0.228	0.052	-0.662	0.230	-0.948	0.343
Kostek et al., 2007 (women) biceps	-0.046	0.195	0.038	-0.428	0.335	-0.237	0.813
Kostek et al., 2007 (men and women) arm	-0.023	0.138	0.019	-0.294	0.248	-0.164	0.869
Kostek et al., 2007 (men) arm	-0.173	0.209	0.044	-0.584	0.237	-0.828	0.408
Kostek et al., 2007 (women) arm	-0.020	0.183	0.033	-0.378	0.339	-0.108	0.914
Krotkiewski, et al., 1979, subcutaneous	-0.751	0.444	0.198	-1.622	0.120	-1.690	0.091
Krotkiewski, et al., 1979, cell fat	-0.735	0.444	0.197	-1.604	0.135	-1.655	0.098
Miura et al., 2009	0.005	0.473	0.223	-0.921	0.932	0.011	0.991
Nickols-Richardson et al. 2007 (concentric), arm	-0.033	0.230	0.053	-0.484	0.418	-0.141	0.887
Nickols-Richardson et al. 2007 (eccentric), arm	0.034	0.243	0.059	-0.442	0.511	0.142	0.887
Nickols-Richardson et al. 2007 (concentric), leg	0.010	0.230	0.053	-0.441	0.460	0.042	0.967
Nickols-Richardson et al. 2007 (eccentric), leg	0.053	0.243	0.059	-0.424	0.530	0.217	0.828
Olson and Edelstein, 1968	-0.504	0.251	0.063	-0.996	-0.012	-2.008	0.045
Orkunoglu-Suer et al., 2008 (women)	0.015	0.079	0.006	-0.140	0.170	0.190	0.849
Orkunoglu-Suer et al., 2008 (men)	0.033	0.101	0.010	-0.164	0.231	0.333	0.739
Ramirez-Campillo et al., 2013, fat mass	0.046	0.380	0.144	-0.699	0.791	0.121	0.903
Ramirez-Campillo et al., 2013, fat percentage	0.000	0.380	0.144	-0.744	0.744	0.000	1.000
Roby, 1962	0.080	0.355	0.126	-0.616	0.777	0.226	0.821
Walts et al., 2008 (men), subcutaneous	0.000	0.155	0.024	-0.305	0.305	0.000	1.000
Walts et al., 2008 (women), subcutaneous	0.101	0.142	0.020	-0.178	0.380	0.707	0.480
Walts et al., 2008 (Caucassians), subcutaneous	0.150	0.132	0.017	-0.110	0.409	1.131	0.258
Walts et al., 2008 (African American), subcutaneous	0.153	0.195	0.038	-0.229	0.535	0.783	0.434
Walts et al., 2008 (men), intermuscular fat	0.119	0.156	0.024	-0.186	0.424	0.767	0.443
Walts et al., 2008 (women), intermuscular fat	-0.096	0.142	0.020	-0.375	0.183	-0.677	0.498
Walts et al., 2008 (Caucassians), intermuscular fat	0.000	0.132	0.017	-0.259	0.259	0.000	1.000
Walts et al., 2008 (African American), intermuscular fat	-0.065	0.195	0.038	-0.446	0.317	-0.332	0.740
Yao et al., 2007	0.122	0.142	0.020	-0.157	0.401	0.857	0.391
	-0.025	0.037	0.001	-0.098	0.048	-0.661	0.508

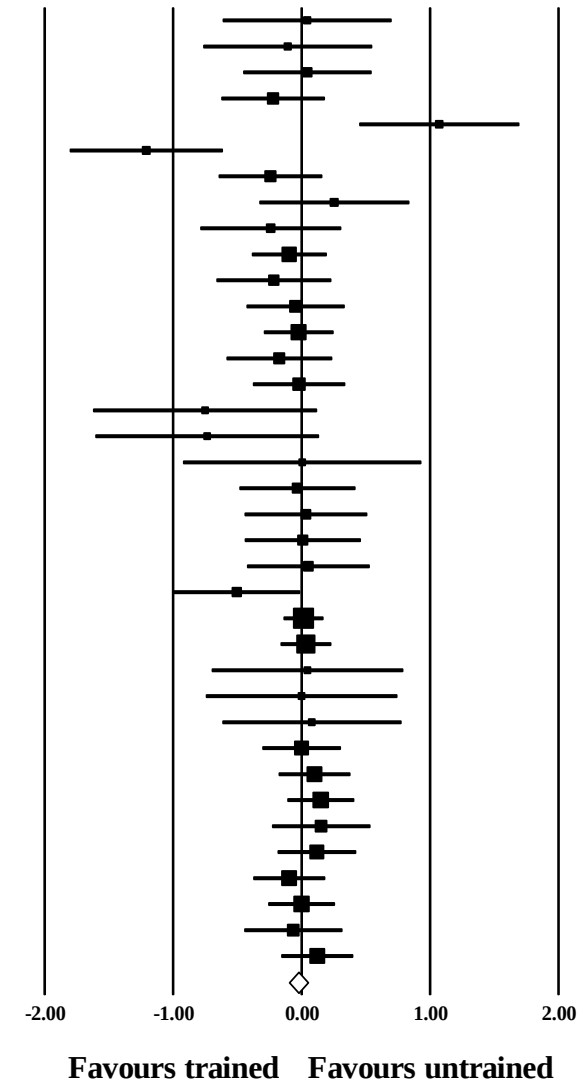


Figure 2. Forest plot for changes in localized fat (spot reduction) in trained compared to untrained limbs. Negative values denote that the trained limb reduced more fat than the untrained limb. Values shown are effect sizes (Hedges's g) with 95% confidence intervals (CI). The size of the plotted squares reflects the statistical weight of each study. The white diamond reflects the overall result.

Table 1. Included studies characteristics.			
Study	Participants	Training	Outcomes
Brinkworth et al., 2004	Healthy physically active men supplemented with bovine colostrum (n=17; age, 21.4 y; height, 179 cm; body mass, 77.8 kg) or whey protein (n=17; age, 23.8 y; height, 179 cm; body mass, 81.5 kg).	8 weeks, 4 sessions per week. Muscle: elbow flexors non-dominant arm. Exercises: dumbbell bicep concentration curls. Velocity: controlled (slower during lengthening). Sets/repetitions/intensity: 6 sets to failure at 80% 1RM. Progressive overload: yes.	Arm skin and subcutaneous fat (cm ² ; MRI).
Devries et al., 2015	30 healthy men (age, 70 y; height, 1.8 m; body mass, 84 kg)	2 weeks, 3 sessions/week. Unilateral leg press and leg extension. Equipment: air-resistance strength machines. Sets, intensity: 3, 30% 1RM until volitional fatigue.	Leg fat mass (g; DEXA).
Hanson et al., 2009	Sedentary (without medical condition) women (n=25; age, 71 y; height, 161 cm; body mass, 75.5 kg; BMI, 29.2 kg·m ⁻²) and men (n=22; age, 71 y; height, 174 cm; body mass, 86.4 kg; BMI, 28.4 kg·m ⁻²).	10 weeks, 3 sessions per week. Knee extensions for the dominant leg (pneumatic [air powered] knee extension machine). Sets: 4-5 (4 for participants >75 y of age and 5 for those <75 y of age). First set: 5 repetitions, 50% 1RM. Second set: 5RM value (initially, 85% of basal 1RM). Third set: 5RM, then a drop-set of 1-2 repetitions until reaching 10 repetitions. Fourth set: 5RM, then a drop-set of 1-2 repetitions until reaching 15 repetitions. Fifth set: 5RM, then a drop-set of 1-2 repetitions until reaching 20 repetitions. Full ROM was required during repetitions. Repetition duration: 2-3 (shortening-lengthening). A seat belt was worn throughout the exercise session, with arms across their chest. Progressive overload was monitored session by session.	Knee extensor subcutaneous fat (cm ² ; computed tomography). Knee extensor intermuscular fat (cm ² ; computed tomography).

Kostek et al., 2007	45 men and 59 women, Caucasian (94%); age, 24.1 y; BMI, 24.2 kg·m ⁻²	12 weeks; 2 sessions per week (45-60 min per session). Progressive, supervised resistance training of their nondominant arm. Exercises: biceps preacher curl, overhead triceps extension, biceps concentration curl, triceps kickback, and standing biceps curl. Dose per exercise: 3 sets of 12 repetitions at 65–75% 1RM (i.e., 12RM). Each contraction involved 2 s for the concentric phase and 2 s for the eccentric phase. A 2-min rest followed each set. At week 5, the number of repetitions was decreased to eight (i.e., 8RM), and then to six (i.e., 6RM) at week 10. Consequently, the exercise intensity at weeks 5 and 10 increased to 75–82% and 83–90% 1RM, respectively. Experienced investigators supervised the training sessions and adjusted the weight accordingly.	Biceps subcutaneous fat (mm; skinfold callipers). Arm subcutaneous fat volume (mL; MRI).
Krotkiewski, et al., 1979	10 women; age, 24-29 y; height, 166.2 cm; body mass, 72-81 kg; body fat, 19-28 kg	5 weeks, performed daily. Three sets of 10 maximal voluntary isokinetic right knee extensions (constant angular velocity of 60 deg·s ⁻¹).	Thigh adipose tissue thickness (cm; ultrasound). Fat cell weight (µg; microscopic method).
Miura et al., 2009	8 women, Japanese, sedentary; ages 21–23 y; height, 157 cm; body mass, 49.4 kg; VO ₂ max, 32.4 mL·min ⁻¹ ·kg ⁻¹	12 weeks, 3 sessions per week (60 min per session). The right or left leg was assigned to cycling at 40% of single leg peak VO ₂ (i.e., below lactate threshold), equivalent to 25.3 W and a heart rate of 90–110 bpm.	Thigh fat cross-sectional area (cm ² ; ultrasound).
Nickols-Richardson et al. 2007	70 women, white (95%); age, 20.2 y; BMI, 22.1 km·m ⁻²	20 weeks; 3 sessions per week. Concentric (or eccentric) slow-velocity (60°·s ⁻¹) isokinetic training of the non-dominant leg and arm. During week 1, one set of 6 repetitions was performed for knee extension and elbow	Arm fat mass (kg; DEXA).

		flexion. From weeks 2 to 5, one set was added each week, so by week 5 participants completed 5 sets of 6 repetitions. From week 6 to week 20, the volume was maintained. Torque output was not controlled during training, but was free to vary (i.e., increase) as participants performed each repetition at maximal volitional effort. Of note, one group of women (n=37) performed concentric training, and the other group (n=33) eccentric training.	Leg fat mass (kg; DEXA).
Olson and Edelstein, 1968	32 boys, with no experience in weight training; age, 14-16 y.	6 weeks, 3 or 5 days per week (half of the participants exercised 5 days a week and the other half exercised 3 days per week; however, data from all the participants was mixed). Right arm curl with dumbbell and triceps extension with dumbbell, for 3 sets of 7RM each exercise (with as many repetitions as possible in the second and third sets). When a sufficient gain in strength allowed seven repetitions to be performed in all three sets, the resistance was increased. There was no warm-up prior to the exercises. The boys did not participate in physical education or in intramural or interscholastic athletics during the study.	Triceps subcutaneous fat (mm; skinfold callipers).
Orkunoglu-Suer et al., 2008	320 women (age, 22.9 y; body mass, 64.7 kg; height, 164.2 cm; BMI, 23.7 kg·m ⁻²) and 197 men (age, 23.9 y; body mass, 78.8 kg; height, 178.5 cm; BMI, 24.7 kg·m ⁻²). All European descents (White).	See: Kostek et al., 2007	Arm subcutaneous fat volume (mm ³ ; MRI).
Ramirez-Campillo et al., 2013	11 physical education students (7 men and 4 women;	12 weeks; 3 sessions per week (80 min per session). Localized muscle endurance resistance training for the non-dominant leg muscles; Subjects completed one set of leg press per session, at 10–30% 1RM (10% during	Leg fat mass (kg; DEXA).

	Latinamerican); age, 23.0 y; BMI, 25.0 kg·m ⁻²	weeks 1–4, 20% during weeks 5–6, and 30% during weeks 7–12). Subjects completed 960–1,200 consecutive repetitions for their set (no rest between repetitions), with 4–5 seconds per repetition.	Leg fat percentage (DEXA).
Roby, 1962	15 male college students; age, 21.1 y	10 weeks, 3 sessions per week. Dominant arm triceps extension, for 3 sets of 10-15 repetitions at 50% 1RM. Overload was applied when participants were able to perform 15 repetitions in all three sets.	Triceps subcutaneous fat (mm; skinfold callipers).
Walts et al., 2008	Men (n=78-82) and women (n=95-98), relatively healthy, physically inactive; age, 63.0 y. Self-reported Caucasians (n=114) or African Americans (n=52).	See: Hanson et al., 2009	Knee extensor subcutaneous fat (cm ² ; computed tomography). Knee extensor intermuscular fat (cm ² ; computed tomography).
Yao et al., 2007	Men (n=46; age, 64.4 y; height, 174 cm; body mass, 84 kg; % body fat, 27.4) and women (n=52; age, 62.7 y; height, 163 cm; body mass, 73.2 kg; % body fat, 38.8)	See: Hanson et al., 2009	Knee extensor intermuscular fat (cm ² ; computed tomography).
Abbreviations (listed alphabetically): 1RM: one repetition maximum; BMI: body mass index; BPM: beats per minute; DEXA: dual-energy x-ray absorptiometry; MRI: magnetic resonance image; ROM: range of motion; VO ₂ : volume of oxygen consumption.			

Table 2. Methodological quality for included studies using the PEDro rating scale.

Study name	Q1	Q2	Q3 ^a	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Total*	Study quality
Brinkworth et al., 2004	1	0	1	1	0	1	1	1	1	1	1	8/10	High
Devries et al., 2015	1	1	1	1	0	0	0	1	1	1	1	7/10	High
Hanson et al., 2009	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Kostek et al., 2007	1	0	1	1	0	1	1	1	1	1	1	8/10	High
Krotkiewski, et al., 1979	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Miura et al., 2009	1	1	1	1	0	0	0	1	1	1	1	7/10	High
Nickols-Richardson et al. 2007	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Olson and Edelstein, 1968	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Orkunoglu-Suer et al., 2008	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Ramirez-Campillo et al., 2013	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Roby, 1962	1	0	1	1	0	0	0	1	1	1	1	6/10	High
Walts et al., 2008	1	0	1	1	0	1	1	1	1	1	1	8/10	High
Yao et al., 2007	1	0	1	1	0	0	0	1	1	1	1	6/10	High

Note: A detailed explanation for each PEDro scale item can be accessed at <https://www.pedro.org.au/english/downloads/pedro-scale>. *: from a possible maximal punctuation of 10. ^a: in the context of this study, Q3 was considered to be attained even if concealed allocation was not reported, since the decision about whether or not to include a person in a trial could not be influenced by knowledge of whether the subject was to receive treatment or not.